

Circuit Simulation Lab:23

Design of Counters

The purpose of this experiment is to introduce the design of Ring Counter. A ring counter is a type of counter composed of flip-flops connected into a shift register, with the output of the last flip-flop fed to the input of the first, making a "circular" or "ring" structure.

CODE:

```
module ring_counter(
input clk,
input rst,
output reg [3:0] q
);

always @(posedge clk)
begin

if(rst)
    q <= 4'b0001;

else
    q <= {q[2:0],q[3]};

end

Endmodule
```

TESTBENCH

```
module tb_ring_counter;

reg clk,rst;
wire [3:0] q;

ring_counter dut(
.clk(clk),
.rst(rst),
.q(q)
);

always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("ring_counter.vcd");  
$dumpvars(0,tb_ring_counter);
```

```
clk=0;  
rst=1;
```

```
#20 rst=0;
```

```
#200;
```

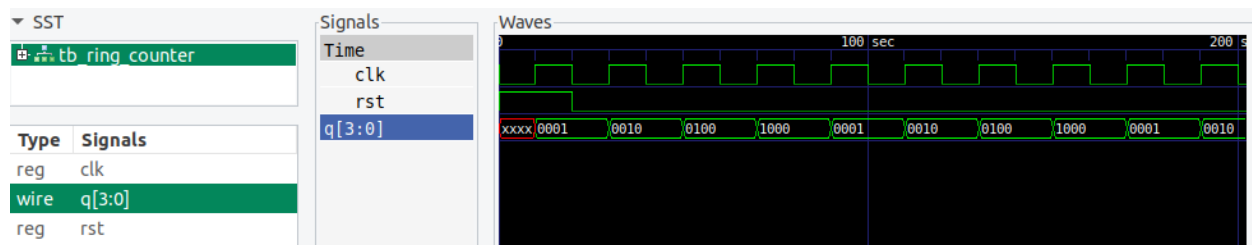
```
$finish;
```

```
end
```

```
initial
```

```
$monitor("time=%0t q=%b", $time, q);
```

```
Endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_23$ iverilog -o out  
ring_counter.v tb_ring_counter.v  
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_23$ vvp out  
VCD info: dumpfile ring_counter.vcd opened for output.  
time=0 q=xxxx  
time=10 q=0001  
time=30 q=0010  
time=50 q=0100  
time=70 q=1000  
time=90 q=0001  
time=110 q=0010  
time=130 q=0100  
time=150 q=1000
```